Problem Statement –

Agentic AI for Intelligent Research Discovery and Literature Intelligence

The Challenge

Researchers, students, and professionals often deal with scattered academic papers, technical articles, and experimental data spread across multiple platforms. Filtering relevant knowledge, summarizing insights, and keeping track of the latest developments is time-consuming and inefficient. The challenge is not just to collect references but to **synthesize, analyze, and generate actionable research insights** in real time.

Objectives

The primary objective of the proposed Agentic AI-powered Research Intelligence Platform is to develop an intelligent research assistant using two collaborative AI agents that automate and enhance the entire academic research lifecycle. The platform aims to continuously discover, collect, filter, classify, and organize research papers, technical articles, patents, and datasets from trusted scholarly sources while monitoring the latest publications in real time. It further seeks to generate actionable research insights by summarizing and comparing research papers, extracting key findings, identifying emerging research trends and knowledge gaps, and producing citation-ready literature reviews with personalized recommendations tailored to researchers' interests. Additionally, the solution enables intelligent knowledge discovery through semantic search, natural language queries, and continuous research monitoring, allowing users to quickly retrieve relevant information and remain informed about recent advancements. To improve research productivity and decision-making, the platform provides an interactive analytics dashboard that visualizes publication trends, topic clusters, citation networks, research timelines, and overall research progress, thereby accelerating literature reviews and supporting high-quality, evidence-based research.